WQD7001 Principles of Data Science - Alternative Assessment 2

Data Science for Urban Sustainability:
Smart Waste Management System (SDG 11)

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Problem Statement

Challenge

Population increase and rapid urbanisation causes waste management challenges, including **inefficient route during waste collection**[1][2].

- Fuel Inefficiency
- Static Routing
- Excess CO_2 Emissions

Data Science Solution

Dynamic and optimised waste collection route using graph neural networks (GNN)[3] and XGBoost prediction.

- Real-time Bin Fill Level
- Dynamic Routing
- Predictive Scheduling

Sustainable Development Goals

11 SUSTAINABLE CITIES AND COMMUNITIES

- **Reduces fuel consumption and CO_2** emission with optimised collection routes
- **Reduces urban pollution and health hazards** from prevented waste overflow

Dataset Illustration

Bin_ID	Timestamp	Fill_Level_Percent	Latitude	Longitude	Demand_kg	Temperature_C	Zone	Priority	Distance_km	Status
BIN001	22/01/2025 08:00	88	2.1896	102.2501	125	32	Zone_A	HIGH	3.2	PENDING
BIN002	22/01/2025 08:00	23	2.1923	102.2534	35	31	Zone_B	LOW	4.5	COLLECTED
BIN003	22/01/2025 08:00	92	2.1887	102.2487	142	33	Zone_A	CRITICAL	2.8	PENDING
BIN004	22/01/2025 08:00	45	2.1945	102.2556	68	30	Zone_C	NORMAL	5.1	COLLECTED
BIN005	22/01/2025 08:00	78	2.191	102.2512	98	32	Zone_A	MEDIUM	3.7	PENDING
BIN006	22/01/2025 08:00	56	2.1875	102.2498	72	31	Zone_A	NORMAL	2.5	COLLECTED
BIN007	22/01/2025 08:00	91	2.1938	102.2523	135	34	Zone_B	CRITICAL	4.2	PENDING
BIN008	22/01/2025 08:00	34	2.1902	102.2545	48	29	Zone_B	LOW	4.8	COLLECTED
BIN009	22/01/2025 08:00	82	2.1919	102.2489	115	33	Zone_A	HIGH	3.1	PENDING
BIN010	22/01/2025 08:00	67	2.1953	102.2512	89	32	Zone_C	MEDIUM	5.3	COLLECTED
BIN011	22/01/2025 08:00	29	2.1884	102.2534	42	30	Zone_A	LOW	3.4	COLLECTED

A **hypothetical** dataset integrating smart bin IoT sensors [3] with vehicle routing problems [5]. The dataset enables **predictive collection scheduling** and **fuel-efficient route planning** for urban waste management.

Future Improvements and Potential Challenges

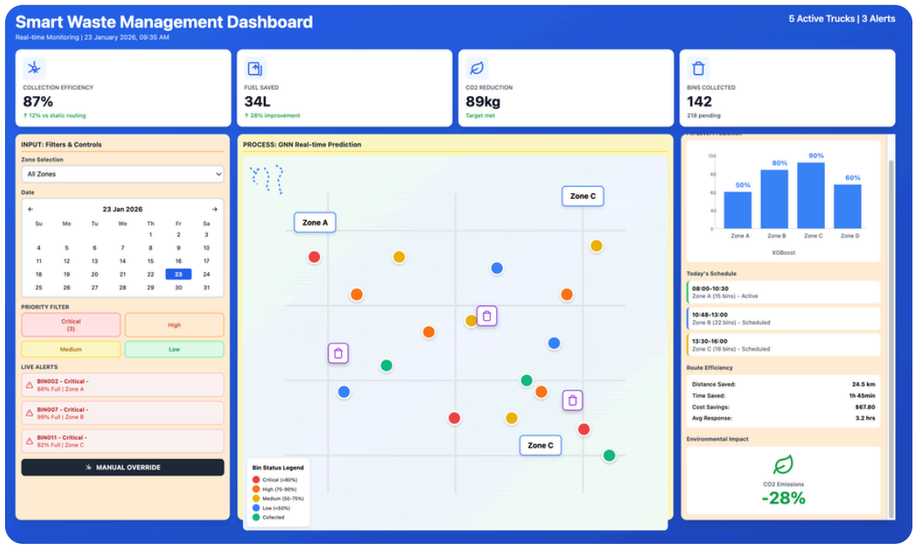
Scalability and Future Improvements

- **Geographic Scale:** Pilot (100 bins) → City-wide (1,000+ bins) → Multi-city deployment
- **API Integration:** Connect to municipal ERP systems for automated billing and reporting
- **Advanced Analytics:** Computer vision for waste composition analysis, supporting recycling optimization

Potential Challenges with Solution

- **Network Coverage:** Use hybrid sensors for rural areas
- **Data Privacy:** Anonymize location data that comply with existing policies
- **Hardware Maintenance:** Predictive maintenance for sensor battery life
- **Cost:** Reduce sensor costs through bulk procurement

Data Product Mock-up



1 Input

- IoT Sensors Bin Level
- GPS Tracking
- Traffic Condition
- Date Selection

2 Process

- GNN forecast optimal routes for collection
- XGBoost predicts bin fill rate
- Anomoly detection to identify unusual fill pattern

3 Actionable Output

- Optimised collection schedule and route
- Route efficiency uplifts
- Reduction of environmental impact

Technologies, Tools and System Functionality

Data Collection

- IoT Sensors
- GPS Modules
- Edge Devices

- IoT ultrasonic sensors identify fill level via LoRaWAN
- GPS track truck locations in real-time
- Edge computing preprocesses sensor data before cloud transmission

Data Processing

- AWS S3
- AWS IoT Core
- PostgreSQL

- AWS IoT Core ingests streaming sensor data
- AWS S3 stores historical data for model training
- Data cleaning and normalization for GNN input

Predictive Analytics

- PyTorch
- XGBoost
- AWS Sagemaker

- PyTorch Geometric: GNN forcast optimal routes for collection using spatial and temporal information
- XGBoost: Regression model to predict bin fill rate with high accuracy

Data Visualisation

- React
- Leaflet
- D3.js

- React dashboard with Leaflet maps for real-time bin visualization
- D3.js charts for predictive analytics
- WebSocket connections for live updates with low latency

References

[1] Sosunova, I., & Porras, J. (2022). IoT-enabled smart waste management systems for smart cities: A systematic review. Ieee Access, 10, 73326-73363.
[2] Munir, M. T., Li, B., & Naqvi, M. (2023). Revolutionizing municipal solid waste management (MSWM) with machine learning as a clean resource: Opportunities, challenges and solutions. Fuel, 348, 128548.
[3] Fey, M., & Lenssen, J. E. (2019). Fast graph representation learning with PyTorch Geometric. arXiv preprint arXiv:1903.02428.
[4] Anitha, R., & Parthiban, A. (2025). AI-IoT-graph synergy for smart waste management: a scalable framework for predictive, resilient, and sustainable urban systems. Frontiers in Sustainability, 6, 1675021.
[5] Joseph, A. (2023). Vehicle routing problem dataset [Data set]. Kaggle.